LAT Pointing Requirements *

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1 Introduction

The basic goal of this project is to determine what the pointing requirements are for various SO LAT science goals, and evaluate whether these goals have been achieved (and ultimately, if not, explore what elements of pointing are limiting us). Some of the framework for how this project will proceed are based on Elisa Russier's talk at the F2F 2026.

The goal can be thought of in a sentence such as:

> **X** must be known to within **a%** to avoid biasing **Y** by more than **b%**.

In the case of pointing, this might look like:

> Pointing must be known to within 5 arcsec to avoid biasing σ_8 by more than 2%.

Where do each of these components come from?

1.1 Science Goals

The enhanced Large Aperture Telescope (LAT) has a number of science goals across cosmological, astrophysical, and particle physics fields, and stemming from a variety of different measurement/analysis channels. Understanding how pointing impacts the science results requires pairing each science goal with its respective analysis method. For example, Abitbol et al. (2025) states the enhanced LAT will place a constraint on the number of relativistic species $\sigma(N_{\text{eff}}) = 0.045$. This constraint is largely set by imprints in the small-scale damping tail of the CMB temperature and polarization anisotropy power spectra. Thus, to determine how pointing impacts this constraint on N_{eff}, one would need to simulate how changes in pointing impact the power spectra (in this case we might predict that pointing errors that express themselves as smoothing of the beam would effective limit/increase uncertainty on the smallest scales of the power spectrum).

1.2 Sources of (LAT) Pointing Error

- Azimuth, Elevation, Corotator encoder offsets
- Axis misalignment between corotator, mirrors, elevation space and the LATR
- Physical elevation sag of mirrors with gravity

Table 1: Summary of Enhanced Science Goals from SO LAT Survey[a]

	Current[b]	**SO 2025–2034**	Using Rubin, DESI, or *Euclid*	Reference
Primordial perturbations				
n_s	0.004	0.002	-	Shandera 2019
$e^{-2\tau}\mathcal{P}(k=0.2\,\mathrm{Mpc}^{-1})$	3%	0.4%	-	Slosar 2019
$f_{\mathrm{NL}}^{\mathrm{local}}$	5	1	✓	Meerburg 2019
Relativistic species				
N_{eff}	0.2	0.045	-	Green 2019
Neutrino mass[c]				
$\sum m_\nu$ (eV, $\sigma(\tau)=0.01$)	0.1	0.03	✓	Dvorkin 2019
$\sum m_\nu$ (eV, $\sigma(\tau)=0.002$)		0.015	✓	
Accelerated expansion				
$\sigma_8(z=1-2)$	7%	1%	✓	Slosar 2019
Galaxy evolution				
η_{feedback}	50–100%	2%	✓	Battaglia 2019
p_{nt}	50–100%	4%	✓	Battaglia 2019
Reionization				
Δz	1.4	0.3	-	Alvarez 2019
τ	0.007	0.0035	-	Alvarez 2019
Cluster catalog	4000	33,000	✓	
AGN catalog	2000	96,000	-	
Galactic science				
Molecular cloud B-fields	10s	> 860	-	SO 2022 GS
$\sigma(\beta_{\mathrm{dust}})$	0.02	0.005	-	SO 2022 GS
Solar System Science				
Distance limit for 5 $M_\oplus$ Planet 9	500 AU	900 AU	✓	Fienga 2020
Asteroid detections		$\sim 10,000$		
Transient detection distance				
Long GRBs, on-axis		1300 Mpc	-	
Low-luminosity GRBs		70–210 Mpc	-	
TDEs, on-axis		670 Mpc	-	

- Tilts of the entire platform
- Thermal fluctuations
- Distortions in focal plane from off-axis mirror design

(Abitbol et al., 2025).

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